

# Authentication & Authorization Document

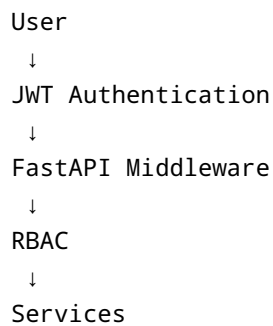
## IntelliPlant AI

### Identity and Access Management

Version: 1.0

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## Authentication Architecture



```
graph TD; User --> JWT[JWT Authentication]; JWT --> FastAPI[FastAPI Middleware]; FastAPI --> RBAC; RBAC --> Services;
```

User  
↓  
JWT Authentication  
↓  
FastAPI Middleware  
↓  
RBAC  
↓  
Services

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## User Roles

### Admin

Full access.

Can:

- Manage users
  - Delete documents
  - Configure system
- 

### Manager

Can:

- View analytics

- Use chat
- View graph

Cannot:

- Delete users
- 

## Engineer

Can:

- Upload documents
  - Query AI
  - Use RCA
- 

## Technician

Read-only access.

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## JWT Flow

Login

↓

Access Token

↓

Middleware

↓

Token Validation

↓

Role Validation

↓

Grant Access

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## Tokens

Access Token

Validity

15 minutes

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Refresh Token

Validity

7 days

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## Password Storage

bcrypt hashing

```
pwd_context.hash(password)
```

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## Middleware

FastAPI Dependency

```
get_current_user()
```

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## RBAC Matrix

| Module              | Admin | Manager | Engineer | Technician |
|---------------------|-------|---------|----------|------------|
| Upload Documents    | ✓     | ✓       | ✓        | ✗          |
| Chat                | ✓     | ✓       | ✓        | ✓          |
| RCA                 | ✓     | ✓       | ✓        | ✗          |
| Compliance          | ✓     | ✓       | ✓        | ✗          |
| Graph Visualization | ✓     | ✓       | ✓        | ✓          |
| User Management     | ✓     | ✗       | ✗        | ✗          |
| Delete Documents    | ✓     | ✗       | ✗        | ✗          |

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## Security Controls

JWT

HTTPS

TLS

Rate limiting

Pydantic validation

bcrypt

Audit logs

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## Audit Logs

Track

- Login
- Upload
- Search
- Agent calls
- Document deletion

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## Future Enhancements

OAuth2

Google Login

Azure AD

SSO

MFA

LDAP

Keycloak

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## Conclusion

JWT + RBAC provide secure access to IntelliPlant AI while ensuring traceability, scalability, and compliance with enterprise security practices.